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1. Introduction

> **What this section covers:** Why the VSI exists, what it measures, and its alternate name.

A pilot can obtain a rough indication of climb or descent rate by watching the angular movement of the altimeter pointer. However, a more accurate indication is needed for tasks such as achieving a specific height loss within a defined time on airways, or establishing a smooth rate of descent on a glide-path during an instrument approach.

The **Vertical Speed Indicator (VSI)** displays *rate of climb or descent* directly. The instrument senses the **rate of change of static pressure**, comparing current static pressure with the static pressure measured `4–6 seconds` earlier. The VSI is also known as the **Rate of Climb and Descent Indicator (RCDI)**.

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2. Principle of Operation

> **What this section covers:** The fundamental physics that drives the VSI needle.

When an aircraft departs from level flight, the ambient static pressure changes. The VSI measures the **pressure difference across a restricted choke/metering unit**.

- **Level flight:** Pressures on both sides of the choke are equal — needle reads zero.
- **Climb:** Capsule pressure falls quickly; choke delays the fall in the case — capsule compresses — pointer indicates climb rate.
- **Descent:** The reverse — capsule expands relative to case — pointer indicates descent rate.

```
flowchart LR
    A["Static Port (current pressure)"] -->|direct feed| B["Capsule (responds instantly)"]
    A -->|via choke / metering unit| C["Instrument Case (responds slowly)"]
    B -- pressure difference --> D["Linkage & Pointer"]
    D --> E["Rate of Climb/Descent Indication"]
```

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3. Construction

> **What this section covers:** The physical components that implement the VSI principle.

The VSI consists of a **capsule within an airtight case**. The capsule connects directly to static and responds immediately. The case also connects to static but through a **restricted choke**, so its pressure changes are delayed. The differential pressure is converted by mechanical linkage to pointer movement.

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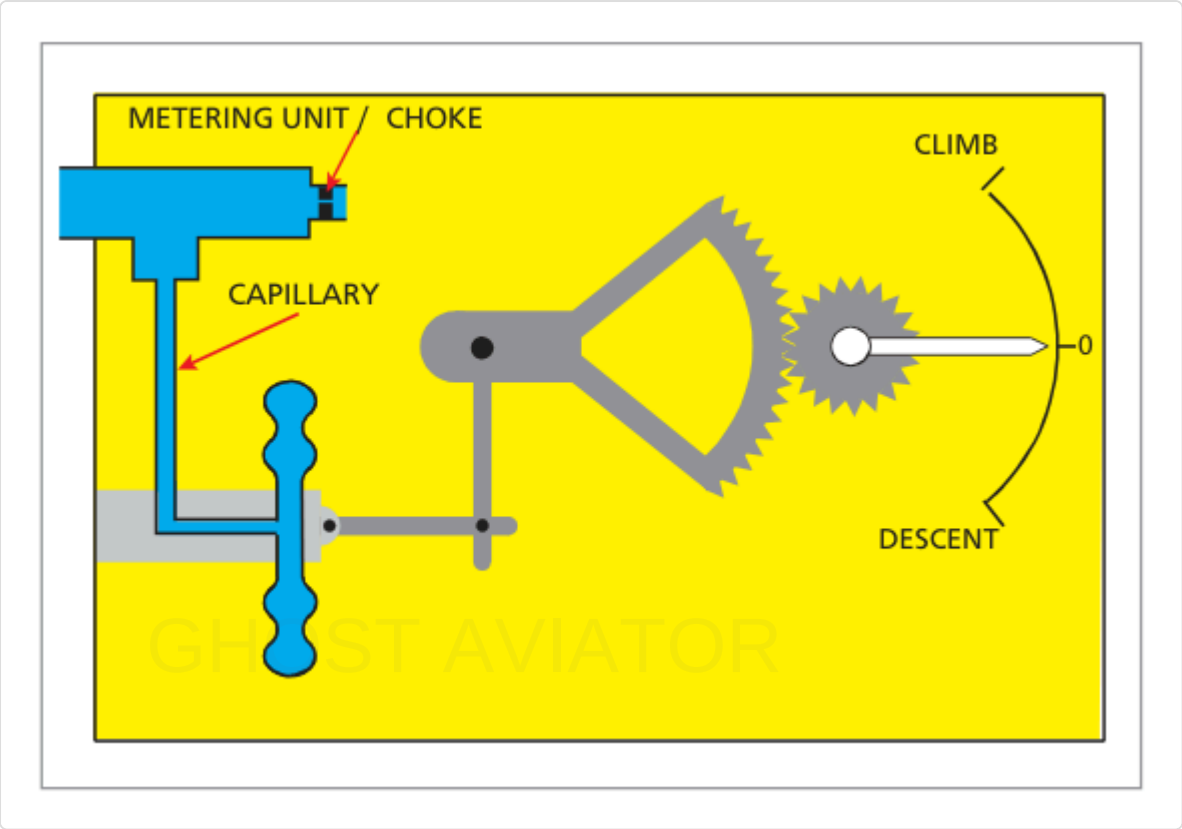


Fig. 6.1 – VSI principle: capsule vs. case separated by metering unit —

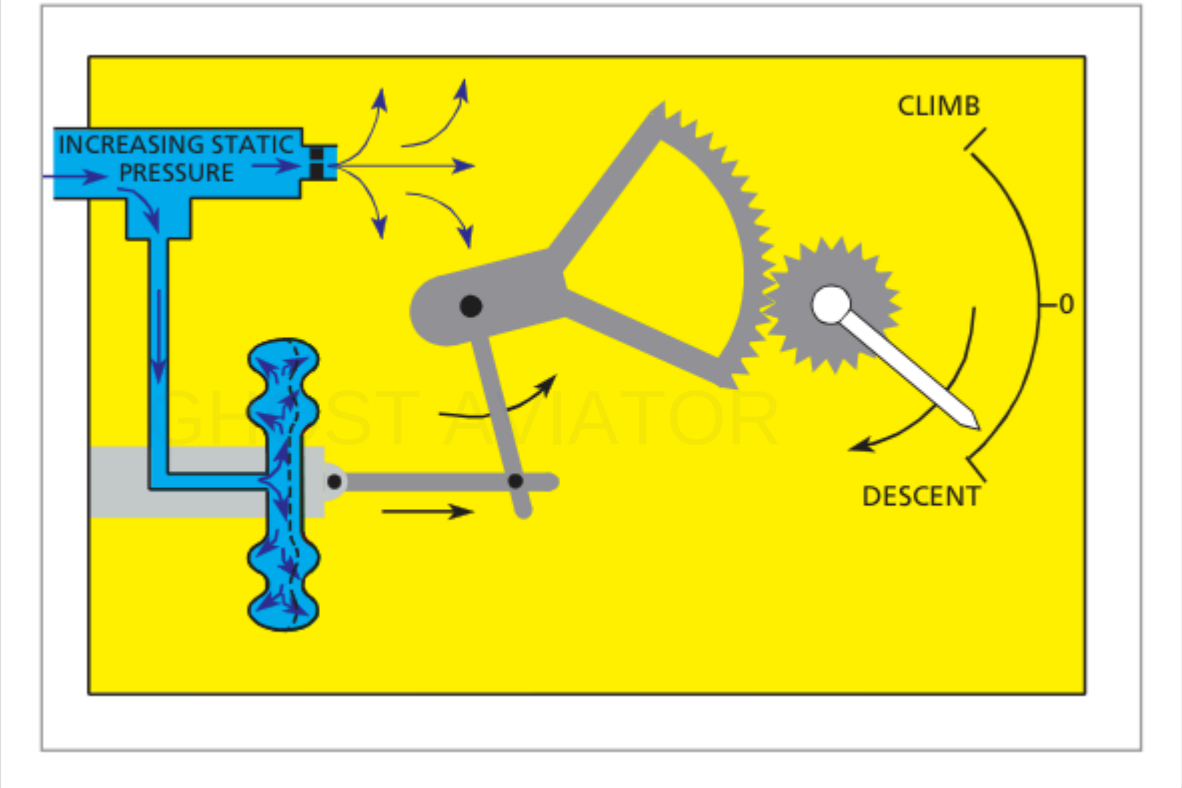


Fig. 6.2 – VSI response during descent (increasing static pressure) —

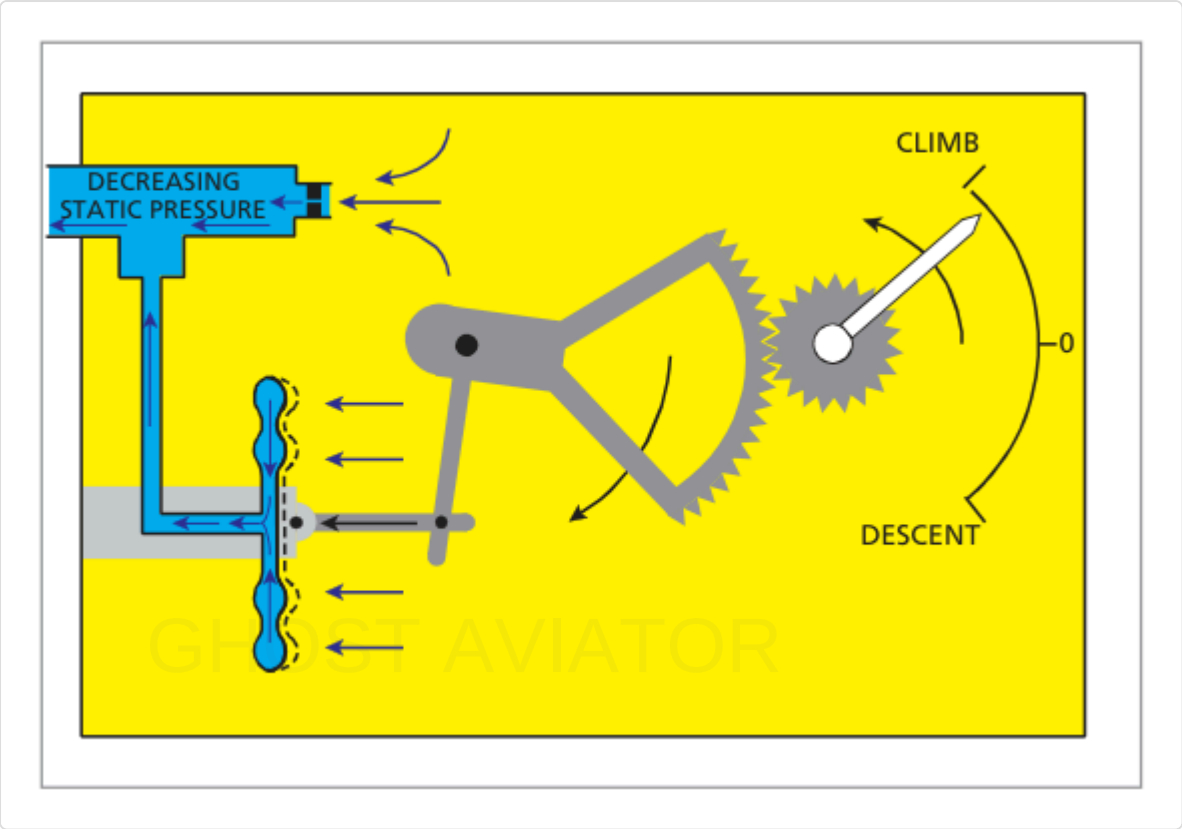


Fig. 6.3 – VSI response during climb (decreasing static pressure) —

4. VSI Metering Unit

> **What this section covers:** Why a simple hole is insufficient, and how altitude compensation is built in.

The basic physics produce an output in *hectopascals per minute*, but pilots need *feet per minute*. The relationship between pressure change and altitude change varies with altitude (more ft/hPa at high altitude). The restrictor uses a **combination of capillary and orifice**:

- **Capillary:** Flow proportional to pressure difference, independent of ambient pressure.
- **Orifice:** Flow varies with ambient air density.

The correct combination produces a near-constant feet-per-minute indication across the operational altitude range.

> **Exam Tip:** The altitude compensation in the VSI is provided by the *capillary and orifice combination* in the metering unit — not a bimetallic temperature sensor.

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5. Display

> **What this section covers:** The VSI face and its features.

The VSI pointer moves clockwise for climbs and anti-clockwise for descents around a centre-zero scale calibrated in **ft/min**. Diaphragm overload stops may be fitted to prevent damage at extreme rates. A zeroing screw is fitted on some instruments.

6. Errors of the VSI

> **What this section covers:** All error types affecting the VSI — critical exam material.

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6.1 Instrument Error

Due to manufacturing imperfections in the linkage and capsule.

6.2 Position (Pressure) Error

If the static source is subject to position error, a rapid airspeed change will cause a false static pressure change. The VSI will indicate a false climb or descent. Most noticeable during **take-off acceleration**.

> **Warning:** A VSI climb indication during the take-off roll caused by position error does not mean the aircraft is climbing — do not react to it until established in the climb.

6.3 Manoeuvre-Induced Error

Short-term static vent pressure fluctuations during attitude changes cause false indications. Additionally, the inertia of the linkage counterbalance weight introduces delays when vertical speed changes during manoeuvres.

6.4 Time Lag

The pointer takes several seconds to settle while a **steady pressure differential** builds across the choke. Similarly, returning to zero on level-off takes time as pressures equalise. Most noticeable after a **prolonged, high-rate climb or descent**.

> **Static Blockage:** Any blockage of the static line or vent causes the needle to **return to zero**. If static is blocked, the ASI, Altimeter, Machmeter, and VSI are all simultaneously affected.

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```
flowchart TD
    E[VSI Errors] --> IE[Instrument Error
Manufacturing imperfections]
    E --> PE[Position Error
Worst during take-off acceleration]
    E --> ME[Manoeuvre-Induced Error
Vent fluctuations + linkage inertia]
    E --> TL[Time Lag
Worst after prolonged high-rate climb/descent]
```

> **Quick Revision – Section 6:** Four errors: Instrument, Position (take-off), Manoeuvre-induced, Time lag (prolonged high-rate). Static blockage → needle to zero, all static instruments affected.

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7. The Instantaneous Vertical Speed Indicator (IVSI)

> **What this section covers:** How the IVSI overcomes lag, and its unique errors.

The IVSI incorporates a **vertical acceleration pump (dashpot / dynamic vane)** that responds immediately to vertical acceleration.

7.1 Mechanism

1. At the start of a descent the dashpot piston **rises immediately**.
2. This temporarily **increases capsule pressure** → capsule expands → pointer indicates descent instantly.
3. As acceleration settles to a constant rate, the piston returns to its resting position; by this time the normal choke-based differential has established and maintains the correct reading.

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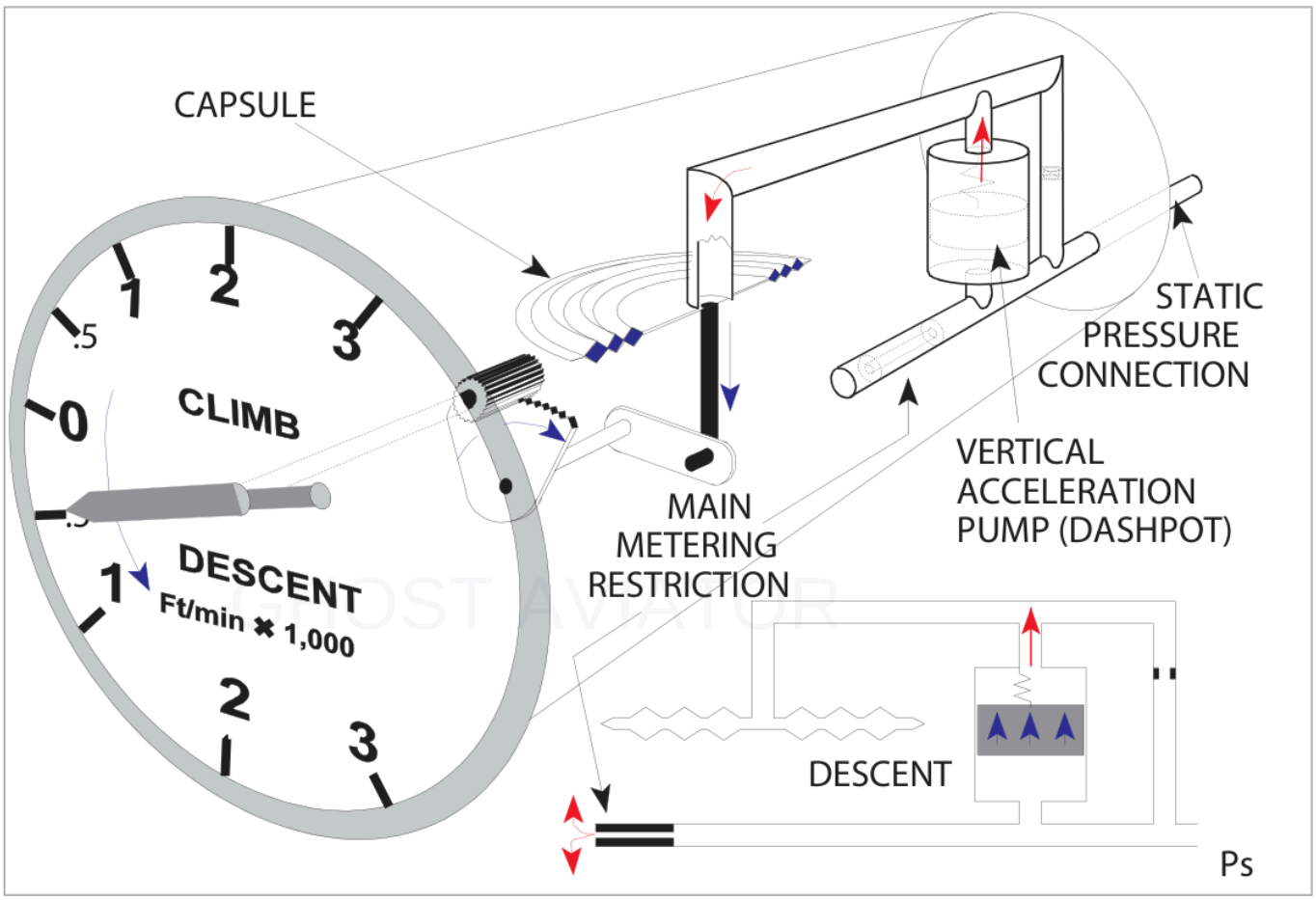


Fig. 6.5 – The Instantaneous Vertical Speed Indicator showing dashpot assembly —

7.2 IVSI-Specific Errors

Turbulence Over-Sensitivity: The highly sensitive dashpot over-reacts to turbulent bumps. Small fluctuations in turbulence should be ignored.

Steep Level Turn — False Climb: Centrifugal force causes the dashpot piston to sink towards the bottom of its cylinder, reducing capsule pressure — the instrument reads a **false climb**.

Exam Tip – IVSI in steep turn: Always shows a false *climb*. Piston sinks → capsule contracts → compass reads climb. This is a classic DGCA trap.

8. Presentation – Linear and Logarithmic Scales

What this section covers: The two dial formats and their respective advantages.

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Feature	Linear Scale	Logarithmic Scale
Graduation spacing	Even throughout	Wide near zero, compressed at high rates
Low-rate readability	Small pointer movement — harder to read	Large pointer movement — easier to read
Range covered	Same as log type	Same as linear type
Preferred use	General purpose	IFR approaches, precision flying

Exam Tip: The logarithmic scale advantage is: larger pointer movement at *low* rates → easier to read. Not a greater range, not instantaneous readings, not a simpler mechanism.

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9. Serviceability Checks

What this section covers: Permitted tolerances and pre-flight/in-flight checks.

On the Ground:

- Read zero, or within: `±200 ft/min` at temperatures `−20°C to +50°C`
- `±300 ft/min` outside this temperature range
- No apparent physical damage

In the Air:

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- Cross-check accuracy against altimeter + stopwatch during steady climb/descent
- Must indicate zero in straight and level flight

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Quick Revision – Chapter 6:

- VSI senses static pressure rate-of-change via capsule/case/choke.
- Metering unit (capillary + orifice) converts hPa/min → ft/min across all altitudes.
- Four errors: Instrument, Position, Manoeuvre-induced, Time lag.
- IVSI: dashpot eliminates lag; over-sensitive in turbulence; false climb in steep turns.
- Logarithmic scale: better resolution at low rates.
- Ground tolerance: `±200 ft/min` (−20 to +50°C), `±300 ft/min` outside.
- Static blockage → needle returns to zero.

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Practice Questions & Detailed Answers

Questions 1–9 with answer key.

Q1. The vertical speed indicator indications may be in error for some seconds after starting or finishing a climb or descent. The error is a result of:

a. a combination of time lag and manoeuvre induced errors

b. a combination of position error and manoeuvre induced errors

c. manoeuvre induced errors only

d. a combination of time lag and instrument error

Correct Answer: **(a)** a combination of time lag and manoeuvre induced errors

Explanation: When starting or finishing a climb/descent both time lag (choke needs seconds to establish new differential) and manoeuvre-induced error (vent fluctuations + counterweight inertia during attitude change) occur simultaneously. See Section 6.

Why the others are wrong:

- **(b)** Position error is specific to sudden airspeed change (take-off), not the general start/finish of a climb.
- **(c)** Manoeuvre-induced alone doesn't explain the prolonged lag seen when levelling off from a long climb.
- **(d)** Instrument error is static and doesn't contribute to transient manoeuvre-related lag.

Instructor's Note: *"Some seconds of error" = time lag + manoeuvre-induced. These two almost always pair together in DGCA questions.*

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Q2. The advantage of having the VSI dial presentation in logarithmic spacing rather than in linear spacing is that:

a. at low rates of climb or descent the pointer movement is much larger and so is more easily read

b. readings are instantaneous

c. a greater range of rates of climb and descent is shown

d. the internal mechanism is simplified by deletion of the calibration choke

Correct Answer: **(a)** at low rates of climb or descent the pointer movement is much larger and so is more easily read

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Explanation: On a logarithmic scale, graduations near zero are widely spaced, so a small rate produces a proportionally larger pointer arc — improving readability at the low rates typical of IFR approaches. See Section 8.

Why the others are wrong:

- **(b)** Instantaneous readings require the IVSI's dashpot — scale type has no effect on lag.
- **(c)** Both scales cover the same range; logarithmic does not extend it.
- **(d)** The choke is still essential regardless of scale presentation.

Instructor's Note: *Logarithmic = better resolution at low rates, NOT greater range or simpler mechanism.*

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Q3. In the IVSI, lag error:

a. is overcome by feeding a sample of static pressure to the case and delaying it to the capsule

b. is overcome by using a special dashpot accelerometer assembly

c. is overcome by the use of logarithmic presentation

d. is only overcome when initiating a climb or descent

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Correct Answer: **(b)** is overcome by using a special dashpot accelerometer assembly

Explanation: The dashpot (vertical acceleration pump) responds instantly to vertical acceleration, providing an immediate pressure signal before the normal choke differential develops. See Section 7.

Why the others are wrong:

- **(a)** Describes the standard VSI — the IVSI adds the dashpot on top of this.
- **(c)** Logarithmic scale is a display feature — no effect on pressure sensing lag.
- **(d)** IVSI overcomes lag in both initiating and stopping a climb/descent.

Instructor's Note: *"Dashpot", "accelerometer", and "dynamic vane" are synonymous IVSI terms — know all three.*

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Q4. Because the VSI measures rates of change of static pressure and not actual values of static pressure, position error:

a. never affects VSI indications

b. may cause errors in the VSI during the take-off run

c. may cause errors in VSI indications whenever airspeed is changed

d. may cause errors in VSI indications whenever airspeed is changed, even if there is no change in position error

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Correct Answer: **(b)** may cause errors in the VSI during the take-off run

Explanation: Position error manifests most strongly when airspeed is changing rapidly. The take-off run is the most significant case of rapid acceleration. The VSI interprets the transient false static pressure as a rate of change. See Section 6.2.

Why the others are wrong:

- **(a)** Position error does affect the VSI.

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- **(c) & (d)** Overstate the effect — not every airspeed change in cruise produces a significant VSI error, only rapid changes create a detectable rate-of-change effect.

Instructor's Note: *The VSI is a rate instrument — position error affects it through the rate of change of position error with changing airspeed, not by its static value alone.*

Q5. When entering a steep turn, an IVSI is likely to show:

a. no change in altitude

b. a climb

c. a descent

d. a slight descent at high airspeed only

Correct Answer: **(b)** a climb

Explanation: In a steep turn, centrifugal force pushes the dashpot piston towards the bottom of its cylinder. This lowers the pressure on the capsule side, which the instrument interprets as a climb (capsule contracts as if static pressure is falling). See Section 7.2.

Why the others are wrong:

- **(a)** The IVSI does react — it shows a false reading, not zero.
- **(c)** A descent indication would require the piston to rise; it sinks under centrifugal force — the opposite effect.
- **(d)** The error occurs in all steep turns regardless of airspeed.

Instructor's Note: *IVSI steep turn = false CLIMB. Memorise: centrifugal → piston sinks → capsule contracts → reads climb.*

Q6. If the static vent becomes blocked during a climb:

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a. the VSI will stop at the rate of climb of the aircraft at the time of blockage

b. the VSI will indicate a decreasing rate of climb

c. the VSI will return to zero

d. the VSI will indicate an increasing rate of climb

Correct Answer: **(c)** the VSI will return to zero

Explanation: With static blocked, no pressure change reaches either the capsule or the case. The existing differential across the choke equalises through the choke itself, collapsing to zero — the needle returns to zero. See Section 6.

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Why the others are wrong:

- **(a)** The instrument does not freeze at the current reading; the differential pressure bleeds away through the choke.
- **(b) & (d)** Neither a decreasing nor increasing indication is sustained — the differential collapses completely.

Instructor's Note: *VSI blockage → zero. Unlike the altimeter (freezes) or ASI (reads take-off speed), the VSI measures differential — so both sides equalise and the reading drops to zero.*

Q7. In conditions of clear air turbulence:

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a. the standard VSI is more sensitive

b. the IVSI is more sensitive

c. both types will react the same

d. the vertical acceleration pump will not be affected

Correct Answer: **(b)** the IVSI is more sensitive

Explanation: The IVSI dashpot responds to every vertical acceleration, including random turbulent bumps. This makes the IVSI pointer flutter considerably in CAT, which is a known limitation. See

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Section 7.2.

Why the others are wrong:

- **(a)** The standard VSI's choke inherently damps short-duration accelerations — it is less reactive to turbulence.
- **(c)** They differ significantly in turbulence response.
- **(d)** The vertical acceleration pump is directly activated by turbulent accelerations.

Instructor's Note: *In severe turbulence, the IVSI pointer may be nearly useless. Use altimeter trend or standard VSI for reliable information.*

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Q8. Change of temperature as an aircraft climbs or descends:

a. will affect VSI readings whenever temperature lapse rate differs from standard conditions

b. is compensated at the metering unit by means of a capillary and orifice

c. has no effect on the VSI as only static pressure is used in this instrument

d. may be allowed for by use of tables or computer

Correct Answer: **(b)** is compensated at the metering unit by means of a capillary and orifice

Explanation: The capillary-and-orifice combination in the metering unit provides built-in compensation for the varying relationship between pressure change and altitude at different altitudes and temperatures — giving a consistent ft/min output. See Section 4.

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Why the others are wrong:

- **(a)** The metering unit compensates for this, so non-standard lapse rates do not significantly degrade calibrated accuracy.
- **(c)** Temperature affects air density and therefore the pressure-altitude relationship — it does matter, but is compensated internally.
- **(d)** No external tables are needed; the metering unit handles it automatically.

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> ***Instructor's Note:*** *Metering unit = capillary + orifice = altitude (density/temperature) compensation. This is the defining design feature of the VSI.*

Q9. Permissible limits of accuracy of the VSI are ..... when ..... within a temperature range of ..... and ..... outside this range.

a. ±250 fpm, on the ground, −20°C to +50°C, ±300 fpm

b. ±200 fpm, at any height, −20°C to +30°C, ±300 fpm

c. ±250 fpm, at any height, −20°C to +50°C, ±300 fpm

d. ±200 fpm, on the ground, −20°C to +50°C, ±300 fpm

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Correct Answer: **(d)** ±200 fpm, on the ground, −20°C to +50°C; ±300 fpm outside this range

Explanation: The serviceability limits are `±200 ft/min` on the ground within the temperature range `−20°C to +50°C`, and `±300 ft/min` outside that range. See Section 9.

Why the others are wrong:

- **(a) & (c)** The tolerance is ±200 fpm, not ±250 fpm — a deliberate numerical distractor.
- **(b)** The temperature range upper limit is +50°C (not +30°C), and the check is on the ground (not at any height).

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> ***Instructor's Note:*** *Learn this sequence: **200 / ground / −20 to +50 / 300**. Tested verbatim in DGCA. The +50 vs +30 confusion is the most common error.*

Master Reference Tables

Parameter	Value	Condition	Section
`Static comparison interval`	4–6 seconds	Pressure window the VSI measures across	§1
`Ground tolerance (normal)`	±200 ft/min	Temperature −20°C to +50°C	§9
`Ground tolerance (extreme temp)`	±300 ft/min	Outside −20°C to +50°C	§9
`IVSI steep turn error`	False CLIMB	Centrifugal force sinks dashpot piston	§7.2
`Static blockage result`	Needle → zero	Differential equalises through choke	§6

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Mnemonics:

- **VSI errors:** "I Put My Tutu" → Instrument, Position, Manoeuvre-induced, Time lag
- **IVSI in steep turn:** "Piston sinks → Reads Climb" (centrifugal sinks piston)
- **Tolerances:** "200 on the ground in the normal range, 300 outside"

Answer Key

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
a	a	b	b	b	c	b	b	d

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